Practical Session 12: Explaining Financial LLM Decisions

Lecture 12 — Hands-On: SHAP by Hand, Faithfulness, and an ECOA Pipeline

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Session Overview

Duration: 1 hour (50 min problems + 10 min debrief)

Agenda:

1. **Recap** (5 min): Shapley formula, faithfulness metrics, the constrained-generation pipeline.
2. **Problem 1** (12 min): Compute Shapley values by hand for a 2-feature credit game; verify the efficiency property.
3. **Problem 2** (12 min): Score sufficiency & comprehensiveness; judge whether an explanation is faithful.
4. **Problem 3** (8 min): Audit an LLM-drafted denial letter for ECOA compliance.
5. **Case Study** (15 min): Run the deterministic SHAP attribution generator in `code/notebooks/12-xai-explainability/`; vary inputs.

Discussion (8 min)

When is a *plausible* explanation worse than no explanation at all?

Recap: Shapley Values and Efficiency

You will need this for Problem 1.

UNDER THE HOOD

$$\phi_i(v) = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!\,(n-|S|-1)!}{n!} \big[v(S \cup \{i\}) - v(S)\big].$$

For $n = 2$, each feature has exactly two orderings, so

$$\phi_1 = \tfrac{1}{2}\big[v(\{1\}) - v(\emptyset)\big] + \tfrac{1}{2}\big[v(\{1,2\}) - v(\{2\})\big].$$

Efficiency (the check):

$$\phi_1 + \phi_2 = v(\{1,2\}) - v(\emptyset) = f(x) - \mathbb{E}[f(X)].$$

The attributions must sum to the gap between prediction and baseline.

Recap: Faithfulness Metrics and the Safe Pipeline

Sufficiency (keep only the explained set e):

$$\mathrm{Suff}(e, f, x) = 1 - |f(x) - f(x_{e(x)})| \quad (\to 1 \text{ is best}).$$

Comprehensiveness (remove the explained set):

$$\mathrm{Comp}(e, f, x) = |f(x) - f(x_{\bar{e}(x)})| \quad (\text{high is good}).$$

The constrained explanation pipeline (for Problem 3)

Decide → attribute (SHAP/IG) → *constrained* LLM translation → compliance filter → human review. The generator may refer **only** to the attribution feed; it must not invent factors or reference protected characteristics (ECOA).

Problem 1: Shapley Values for a Credit Game

A credit model scores an applicant with two features: x_1 = debt-to-income, x_2 = revenue trend. The value function $v(S)$ = model's P(creditworthy) when only the features in S are known:

Coalition S	$v(S)$
$\emptyset$ (baseline)	0.50
$\{1\}$ (DTI only)	0.32
$\{2\}$ (revenue only)	0.41
$\{1,2\}$ (both)	0.14

Task A. Compute ϕ_1 and ϕ_2 from the $n = 2$ formula.
Task B. Verify the efficiency property $\phi_1 + \phi_2 = v(\{1,2\}) - v(\emptyset)$.
Task C. Which feature is the stronger driver of the rejection? In an ECOA notice, which reason is listed first?

Problem 2: Is This Explanation Faithful?

A text classifier outputs $f(x) = 0.14$ for a loan application (baseline 0.50). An XAI method claims the explained set $e(x) = \{$"struggled", "declining", "down 30%"$\}$ accounts for the decision. You run two ablations:

- Keep *only* $e(x)$ (mask all other tokens): $f(x_{e(x)}) = 0.19$.
- Remove *only* $e(x)$ (keep the rest): $f(x_{\bar{e}(x)}) = 0.44$.

Task A. Compute the sufficiency $\text{Suff}(e, f, x)$.
Task B. Compute the comprehensiveness $\text{Comp}(e, f, x)$.
Task C. Is this explanation *faithful*? Interpret both numbers — what would a low sufficiency *and* a low comprehensiveness have told you?
Task D. The attention map for the same input highlights "restaurant" and "market" instead. After ablating those two tokens, f is unchanged at 0.14. What does that imply about the attention explanation?

Problem 3: Audit an LLM-Drafted Denial Letter

An LLM was given this attribution feed (DTI -0.18, revenue -0.12, cash reserve -0.07) and produced the draft below. **Find every compliance defect** against ECOA / Reg. B.

Draft denial letter

"Dear Applicant, your application was declined. Your overall credit profile did not meet our criteria. We also note that applicants in your area and of your demographic tend to default more often, and your business outlook seemed weak. We wish you the best."

Task. Identify which of the following the draft violates and why:

1. Vagueness (specificity requirement).
2. Introduction of a factor *not* in the attribution feed.
3. Reference to a protected characteristic.
4. Missing required disclosures (statement-of-reasons / consumer-report rights).

Rewrite the first sentence to be *specific and compliant* using the feed.

Case Study: Deterministic SHAP Attribution Figure

Goal: reproduce the token-level SHAP attribution chart from the chapter's credit example, then modify it.

Code: `code/notebooks/12-xai-explainability/gen_shap_attribution.py`

Why deterministic? The script hard-codes the documented attribution values from the example — it depends on *no* live model or data, so every student reproduces an identical figure. This is exactly the *reproducibility* property auditors require.

Prerequisites

```
pip install matplotlib
```

Run: `python3 gen_shap_attribution.py` (writes a PDF into the book's `figures/` directory and prints the sum of shown tokens).

Case Study: The Code

```
TOKENS = [
    ("struggled",   -0.089),
    ("declining",   -0.073),
    ("down 30%",    -0.061),
    ("competitive", -0.047),
    ("restaurant",  -0.038),
    ("pandemic",    -0.029),
    ("operates",    +0.012),
]

# sort most-negative at the bottom for a waterfall read
items  = sorted(TOKENS, key=lambda t: t[1])
labels = [t[0] for t in items]
values = [t[1] for t in items]
colors = ["#d62728" if v < 0 else "#2ca02c" for v in values]

fig, ax = plt.subplots(figsize=(7.2, 4.2))
ax.barh(range(len(values)), values, color=colors)
ax.axvline(0.0, color="black", linewidth=0.8)
ax.set_xlabel("SHAP value (contribution to P(creditworthy))")
fig.savefig(OUT, bbox_inches="tight")
print("sum of shown tokens =", round(sum(values), 3))
```

Baseline 0.50 $\rightarrow$ model output 0.14; total gap $= -0.36$.

Case Study: Diagnostic Questions

Run the script, then answer:

Q1. What is the printed sum of the seven shown tokens? Compare it to the full prediction gap $0.14 - 0.50 = -0.36$. Why is the sum of the *shown* tokens not exactly -0.36? (Hint: *completeness*.)

Q2. Compute the SHAP *completeness* ratio $\sum_{i \in e} \phi_i / (f(x) - \mathbb{E}[f(X)])$ for the seven tokens. What fraction of the decision do they explain?

Q3. Change "operates" from $+0.012$ to -0.030. Does the colour of its bar change? What would such a sign flip imply for the adverse-action narrative?

Q4. The figure is *deterministic*. Contrast this with running *KernelSHAP* or *LIME* live on the model. Which property required for audit does the deterministic version guarantee that a live, unseeded run does not?

Discussion: When Is a Plausible Explanation Dangerous?

For each scenario, decide whether the explanation is safe to ship, and what extra check you would require.

1. A retail-facing chatbot generates a fluent paragraph explaining a loan rejection. It was *not* given the upstream SHAP feed.
2. An attention heat-map highlights tokens an expert agrees are relevant, but ablating them does not change the prediction.
3. A robo-adviser's suitability narrative cites a 9% expected return that does not appear in the product's Key Information Document.

Frame your answer around:

faithfulness vs. plausibility, the separation-of-concerns pipeline, protected characteristics, and reproducibility for audit.

Solution: Problem 1 — Shapley Values

Task A.

$$\phi_1 = \tfrac{1}{2}[v(\{1\}) - v(\varnothing)] + \tfrac{1}{2}[v(\{1,2\}) - v(\{2\})] = \tfrac{1}{2}(0.32 - 0.50) + \tfrac{1}{2}(0.14 - 0.41)$$

$$= \tfrac{1}{2}(-0.18) + \tfrac{1}{2}(-0.27) = -0.090 - 0.135 = -\mathbf{0.225}.$$

$$\phi_2 = \tfrac{1}{2}(0.41 - 0.50) + \tfrac{1}{2}(0.14 - 0.32) = \tfrac{1}{2}(-0.09) + \tfrac{1}{2}(-0.18) = -\mathbf{0.135}.$$

Task B (efficiency).
$\phi_1 + \phi_2 = -0.225 - 0.135 = -0.360 = v(\{1,2\}) - v(\varnothing) = 0.14 - 0.50.$ ✓

Task C. $|\phi_1| > |\phi_2|$: **debt-to-income** is the stronger driver $\Rightarrow$ listed first in the ECOA notice.

Solution: Problem 2 — Faithfulness

Task A (sufficiency). Suff $= 1 - |0.14 - 0.19| = 1 - 0.05 = \mathbf{0.95}$ — high: keeping only $e(x)$ nearly reproduces the prediction.

Task B (comprehensiveness). Comp $= |0.14 - 0.44| = \mathbf{0.30}$ — large: removing $e(x)$ moves the prediction most of the way back toward the 0.50 baseline.

Task C. High sufficiency *and* high comprehensiveness $\Rightarrow$ the explanation is **faithful**: the three tokens both suffice for and are necessary to the decision. Low sufficiency would mean $e(x)$ omits real drivers; low comprehensiveness would mean the model routes around $e(x)$.

Task D. The attention explanation **fails comprehensiveness** (ablating its tokens leaves $f = 0.14$ unchanged). It is *plausible* but not *faithful* — not audit-grade.

Solution: Problem 3 — Denial-Letter Audit

Defects in the draft:

1. **Vagueness:** "credit profile did not meet our criteria" — no specific, quantified reasons. Violates Reg. B specificity.
2. **Unlisted factor:** "business outlook seemed weak" is not in the feed; the generator hallucinated it.
3. **Protected characteristic:** "applicants … of your demographic" — prohibited basis under ECOA. Severe.
4. **Missing disclosures:** no statement-of-reasons right, no free-consumer-report-copy notice.

Compliant rewrite of the first sentence:

"Your application was declined primarily because your debt-to-income ratio of 54% exceeded our maximum of 45%, your revenue declined 28% year-on-year, and your cash reserves covered under one month of operating expenses."

Specific, quantified, drawn *only* from the feed, no protected factors.

Wrap-Up: Key Takeaways

From Problem 1: Shapley values are marginal contributions averaged over orderings; the efficiency property is your built-in correctness check, and $|\phi_i|$ orders the ECOA reasons.

From Problem 2: Faithfulness = sufficiency *and* comprehensiveness. Attention can be plausible yet fail comprehensiveness — not audit-grade.

From Problem 3: The dangerous failures are hallucinated factors and protected-characteristic leakage. Constrain the generator + add a compliance filter + human review.

From the Case Study: Deterministic, seeded explanations are reproducible — the property audits demand and live LIME/KernelSHAP runs do not guarantee.

Next practical

Privacy and local models: running explainable pipelines on-premises when client data cannot leave the institution.

Appendix

Stretch Problems

Extra / optional material — beyond the core lecture.

Appendix: Stretch Problem — Counterfactual Search

A scoring model uses two structured features, DTI d and cash-reserve months c. The decision boundary (approve iff score ≥ 0.5) is score $= 0.5 - 1.2(d - 0.45) + 0.1(c - 3)$, where d is a fraction.

An applicant has $d = 0.54$, $c = 0.8$.

1. Compute the current score. Is the applicant approved?
2. Find the minimal *single-feature* change in d (holding c fixed) that flips the decision. Express it as an actionable counterfactual.
3. Repeat for c. Which counterfactual is the smaller *relative* change?
4. Why might the d-counterfactual be preferred even if the c-change is numerically smaller? (Hint: feasibility & cost-reflecting distance.)

Sketch: score $= 0.5 - 1.2(0.09) + 0.1(-2.2) = 0.5 - 0.108 - 0.22 = 0.172 < 0.5 \Rightarrow$ declined. Approval needs the linear term ≥ 0: e.g. $d \leq 0.45 - (0.1(c - 3))/1.2$. Articulate the resulting threshold as a plain counterfactual.